METRIC
MIL-PRF-18458D
7 July 2022
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PERFORMANCE SPECIFICATION

GREASE, WIRE ROPE AND EXPOSED GEAR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers corrosion resistant, externally applied, grease for use on wire ropes and exposed gears of hoists, winches, cranes, shovels, dredges, and other similar equipment.

1.2 Classification. The following two types of grease are as specified (see 6.2):

a. Type I – For general application.

b. Type II – Environmentally preferable, for applications where potential discharge to the environment is greater.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910 Subpart Z - Toxic and Hazardous Substances

(Copies of this document are available online at https://ecfr.federalregister.gov/.)

NATIONAL TOXICOLOGY PROGRAM (NTP)

Report on Carcinogens

(Copies of this document are available online at www.niehs.nih.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering hard copy via the Naval Logistics Library [NLL] at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org.)

ASTM INTERNATIONAL

ASTM A1008/A1008M	-	Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable
ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D217	-	Standard Test Methods for Cone Penetration of Lubricating Grease
ASTM D1748	-	Standard Test Method for Rust Protection by Metal Preservatives in the Humidity Cabinet
ASTM D2596	-	Standard Test Method for Measurement of Extreme-Pressure Properties of Lubricating Grease (Four-Ball Method)
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D5864	-	Standard Test Method for Determining Aerobic Aquatic Biodegradation of Lubricants or Their Components
ASTM D6139	-	Standard Test Method for Determining the Aerobic Aquatic Biodegradation of Lubricants or Their Components Using the Gledhill Shake Flask

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ASTM D6184 - Standard Test Method for Oil Separation from Lubricating Grease (Conical Sieve Method)

ASTM G154 - Standard Practice for Operating Fluorescent Ultraviolet (UV) Lamp Apparatus for Exposure of Nonmetallic Materials

(Copies of these documents are available online at www.astm.org.)

ORGANIZATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT (OECD)

OECD 202 - OECD Guidelines for Testing of Chemicals, Test No. 202: Daphnia sp. Acute Immobilization Test

OECD 301B - OECD Guidelines for Testing of Chemicals, Test No. 301: Ready Biodegradability, Method 301B: CO_2 Evolution (Modified Sturm Test)

OECD 301F - OECD Guidelines for Testing of Chemicals, Test No. 301: Ready Biodegradability, Method 301F: Manometric Respirometry

(Copies of these documents are available online at https://www.oecd-ilibrary.org/.)

INTERNATIONAL AGENCY FOR RESEARCH ON CANCER (IARC)

IARC Monographs on the Identification of Carcinogenic Hazards to Humans

(Copies of this document are available online at http://publications.iarc.fr/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The grease furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material. The grease shall consist of a liquid lubricant and gelling agent that, when formulated, results in a compound meeting this specification. The grease shall not contain additives of the solid type. Both types shall be of such material as to not require additional container labeling in accordance with 29 CFR 1910 Subpart Z.

3.3 Toxicity and prohibited materials.

3.3.1 Toxicity. When evaluated in accordance with 4.5.9, the grease shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.9, 6.1, and 6.5).

3.3.2 Prohibited materials. The product shall not contain any 2,6-di-t-butylphenol, chlorinated compounds, or hydrolyzable chlorine derivatives. All derivatives or salts of barium, strontium, lead, and chromium, including chromates or dichromates, are also prohibited. The grease shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 (see 4.5.10).

3.4 Recycled, recovered, environmentally preferable materials, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Chemical and physical properties. Type I and type II grease shall have the chemical and physical properties shown in table I. Type II grease shall also meet the environmental requirements of table II.

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TABLE I. Chemical and physical requirements and test methods for type I and type II grease.

Characteristics	Requirement	Applicable standard	Verification	Qualification	Conformance
Penetration, worked (60 double strokes)	200 to 350 range	ASTM D217	4.5	x	x
Oil separation (%)	10 maximum	ASTM D6184 1/	4.5	x	x
Load wear index, kilogram force	30 minimum	ASTM D2596	4.5	x	x
Salt spray	Pass	ASTM B117	4.5.2	x	-
Accelerated weathering	Pass	ASTM G154	4.5.3	x	-
Low temperature flexibility	Pass	-	4.5.4	x	x
Volatile matter (%)	2.0 maximum	-	4.5.5	x	x
Adhesiveness (%)	95 minimum	-	4.5.6	x	x
Toxicity	3.3.1	-	4.5.9	x	-
Prohibited materials	3.3.2	-	4.5.10	x	-
Compatibility	3.6	-	4.5.11	x	-

NOTE:

1/ Test shall be conducted at 65.5°C (150°F) for 50 hours duration.

TABLE II. Additional requirements and test methods for type II grease.

Characteristics	Requirement	Applicable standard	Verification	Qualification	Conformance
Aquatic biodegradability, (%)	≥60% biodegradation after 28 days	OECD 301F 1/	4.5.7	x	-
Aquatic acute toxicity, mg/l	1,000 minimum	OECD 202	4.5.8	x	-

NOTE:

1/ Acceptable alternatives to the test method and limit are ≥60% in 28 days by ASTM D5864, ASTM D6139, or OECD 301B test methods.

3.6 Compatibility. The grease shall be compatible with all currently qualified greases when tested in accordance with 4.5.11. Immediately after mixing, and after undisturbed storage for 7 days, the mixture shall not form resinous gums, sludge, or insoluble solid materials and shall comply with the worked penetration requirement (see table I). In the event of failure, the currently qualified grease (unmixed with other greases) shall be tested to verify that it meets the worked penetration requirement under the conditions specified in 4.5.11. If the currently qualified sample fails to meet the worked penetration requirement, a fresh sample shall be obtained and the compatibility test repeated.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification tests shall be conducted at a laboratory acceptable to Naval Sea Systems Command (NAVSEA). Qualification inspection for type I grease shall include the examination of 4.4 and all of the tests specified in table I and 4.5 (except 4.5.7 and 4.5.8). Qualification inspection for type II grease shall include the examination of 4.4 and all of the tests specified in table I, table II, and 4.5.

4.2.1 Sampling. Qualification test samples shall consist of a minimum of 15.9 kilograms (35 pounds) of grease.

4.2.2 Change approval. Changes in construction, formulation, materials, design, or manufacturing process of qualified products may require requalification. Unless otherwise specified by the qualifying activity, changes require written approval by NAVSEA.

4.2.3 Requalification. Unless otherwise approved by NAVSEA, requalification shall consist of all the tests specified in the qualification test column of tables I and II to satisfy the requirements outlined in section 3 (see 6.3).

4.3 Conformance inspection. Conformance inspection for type I and type II grease shall include the examination of 4.4 and all of the tests specified in table I, except the salt spray (see 4.5.2) and accelerated weathering (see 4.5.3) tests need not be run. Samples for conformance inspection shall be as specified in 4.3.1. Failure of the samples in any of these tests shall be cause for rejection of the lot (see 6.4.1).

4.3.1 Sampling for tests. Two 2.95-kilogram (6.5-pound) containers of grease shall be selected and taken at random from each filled lot (see 6.4.1). For users who obtain grease in large containers, samples shall be selected in accordance with ASTM D4057. One sample shall be used for tests and the other shall be retained. If any test result does not comply with the stated requirement then the test in question shall be repeated on the retained sample. The lot shall be rejected if both samples fail the test.

4.3.2 Sampling for examination of filled containers. A random sample of filled containers shall be drawn from each lot in accordance with ANSI/ASQ Z1.4 to verify compliance with this specification regarding fill, closure, marking, and other requirements not involving test.

4.3.3 Examination of filled containers. Samples of filled containers selected from each lot in accordance with MIL-STD-1916 as specified (see 6.2) shall be examined to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests. Containers shall be examined for defects of the container and the closure, for evidence of leakage, and for unacceptable markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or containing less product than stated on the container (under required fill) shall be cause for rejection of the container. If the number of defective containers in any sample exceeds the acceptable number for the appropriate sampling plan specified (see 6.2), this shall be cause for rejection of the lot represented by the sample.

4.4 Examination. The grease shall be examined for compliance with the requirements specified in 3.2.

4.5 Test methods. Testing shall be performed in accordance with ASTM test methods, applicable standards, or applicable test paragraphs as shown in tables I and II.

4.5.1 Preparation of specimens for salt spray and accelerated weathering tests. Three specimen panels shall be prepared for each test. For salt spray test, see 4.5.2; for accelerated weathering test, see 4.5.3.

4.5.1.1 Panel composition and size. Steel panels shall be made of commercial steel (CS) type in accordance with ASTM A1008/A1008M. The panels shall be 5.08 by 10.16 by 0.318 centimeters (2 by 4 by 0.125 inches) in accordance with those panels specified in ASTM D1748.

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4.5.1.2 Panel finishing and cleaning. The panel test surfaces shall be finished and cleaned as follows:

a. After rounding the edges of the panel and reaming out the holes used for suspension, wipe the surfaces as clean as possible using solvent-soaked rags.

b. Scrub the panel with clean cotton or surgical gauze swab in a beaker of hot nonaromatic petroleum solvent such as heptane.

c. Rinse in warm (21.11 to 26.67 °C [70 to 80 °F]) isopropyl alcohol for at least 10 seconds and air-dry the panel. If required, use a vacuum desiccator and set it aside for later usage.

d. Buff the panel test surface with 240 grit aluminum oxide abrasive with plain or paper backing so as to produce a surface finish of 0.254 to 0.508 micrometer (10 to 20 micro inches) root mean square. The final abrasion marks shall be in the direction parallel to the length of the panel.

e. Wipe off superficial dust from the abrasion operation using a clean, dry absorbent tissue or cloth.

f. Scrub abraded face of panel thoroughly with a clean, lint-free cloth until there is no dark stain on a clean section. Surgical gauze held in a blotter holder is convenient for this operation.

g. Spray the panel with hot, nonaromatic hydrocarbon solvent (e.g., heptane) using a wash bottle or a spray gun. Hold the panel in a rack at 20 degrees from the vertical. Direct the spray vertically downward on the panel, flushing the test surface progressively downward. Spray the test surface, back face, and finally the test surface again.

h. Finally rinse the panel in fresh boiling isopropyl alcohol. Immerse the panel for at least 10 seconds to warm up to the solvent temperature prior to withdrawal. Rinsing operations shall be conducted under a hood for safety.

i. Permit the panel to dry and preserve it in a desiccator. Use the panel on the day of preparation.

4.5.1.2.1 Precautions in cleaning and finishing procedure.

a. In all stages of treatment, handling of equipment with bare hands shall be avoided. The panels shall be handled with hooks, forceps, or similar devices, care being taken to prevent contact of the panels with contaminated surfaces during the cleaning procedure.

b. The utensils and cloths used in the preparation of the panels shall be clean and free of contamination. Solvents shall be clean and renewed frequently.

4.5.1.3 Coating of test panels. The greases shall be applied to the panels with a spatula or other approved tool, taking care to cover all surfaces with a uniform thickness of grease. 1 milliliter of sample for coating shall be applied to the forward-facing test surface of the panel. The edges and back side of the panel shall be liberally coated with additional test grease to help prevent excessive corrosion.

4.5.2 Salt spray test. The grease-coated panels shall be exposed to a 20 percent sodium chloride solution spray in a salt spray cabinet in accordance with ASTM B117 for 10 days ±1 hour. The panels shall be removed from the cabinet and shall be rinsed in warm water to remove any salt residue. The panels shall be rinsed in hot, nearly boiling, nonaromatic hydrocarbon solvent (e.g., heptane) to remove the grease coating. The panels shall be examined visually. Any corrosion appearing on the outer 0.635 centimeter (0.25 inch) of the panel shall be disregarded. The panels shall be considered passing if none of the panels show more than three corrosion spots, each of which is not greater than 1 millimeter (mm) in diameter.

4.5.3 Accelerated weathering test. Three specimens prepared as specified in 4.5.1.3 shall be exposed in a fluorescent ultraviolet (UV) lamp apparatus in accordance with ASTM G154 for 10 days ±1 hour. Black panel temperature shall be 48.9±1.7 °C (120±3 °F) during the light-on, without-water-spray period of the test cycle. Apparatus shall be operated 18 hours for each of the testing days. At the end of the exposure period, the panels shall be removed from the test equipment. The panels shall be rinsed in hot, boiling, nonaromatic hydrocarbon solvent and shall be examined as specified in 4.5.2. The panels shall be considered passing if none of the panels show more than three corrosion spots, each of which is not greater than 1 millimeter in diameter.

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4.5.4 Low temperature flexibility.

4.5.4.1 Apparatus and material. Apparatus and material shall be as follows:

a. Brass strip, approximately 10.16 by 1.27 by 0.01778 centimeters (4 by 0.5 by 0.007 inches).

b. Strip coating apparatus as shown on figure 1.

c. Mandrel 3.81 centimeters (1.5 inches) in diameter.

d. Constant temperature box maintained at -6.67 °C (20 °F).

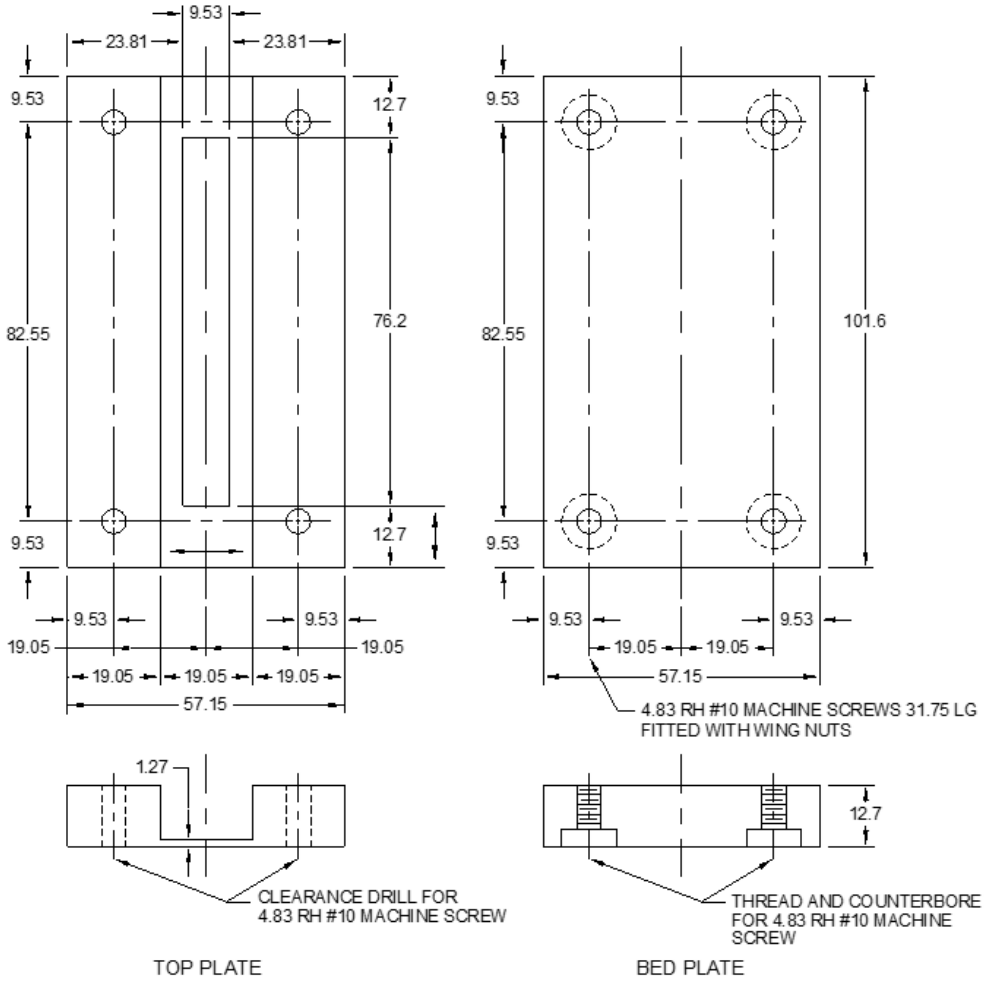


FIGURE 1. Strip coating apparatus for flexibility test.

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4.5.4.2 Procedure. The brass strip shall be cleaned with 240-grit abrasive paper the long way of the strip. The strip shall be washed with nonaromatic petroleum solvent and shall be dried. The strip shall be clamped between the top plate and bottom plate in the stripcoating apparatus. The strip shall be coated by spreading the grease over the surface of the thickness to 0.127 centimeter (0.05 inch) by drawing a knife-edge across the top of the grease well. The strip shall be removed from the coating apparatus. The strip and mandrel shall be placed in the constant temperature box for 1 hour. With the bare metal next to the mandrel, the strip shall be bent in a steady, slow motion for 180 degrees around the mandrel in the box. The film shall not be touched before or during the test and the bending shall be completed in not more than 2 seconds. The bent strip shall be removed from the box and the coating shall be examined for cracks. The bent strip shall be checked for retained flexibility. The material shall be considered as meeting the requirement if the grease surface does not crack during this procedure.

4.5.5 Volatile matter. Approximately 2 grams (0.07 ounce) of the grease shall be weighed to the nearest milligram in a tared aluminum foil dish 5.0 centimeters (2.0 inches) in diameter. The dish shall be placed in an explosion–proof, gravity convection oven maintained at 105 to 110 °C (221 to 230 °F) for a period of 3 hours. The volatile contents are determined from the resultant weight loss.

4.5.6 Adhesiveness.

4.5.6.1 Apparatus. Apparatus shall be as follows:

a. Disc and shaft as shown on figure 2.

b. Vertical shaft and motor assembly capable of rotating disc at 150±10 revolutions per minute.

c. Oven maintained at 66.1±1.1 °C (151±2 °F).

d. Drip shield approved for catching grease flung centrifugally from spinning disc.

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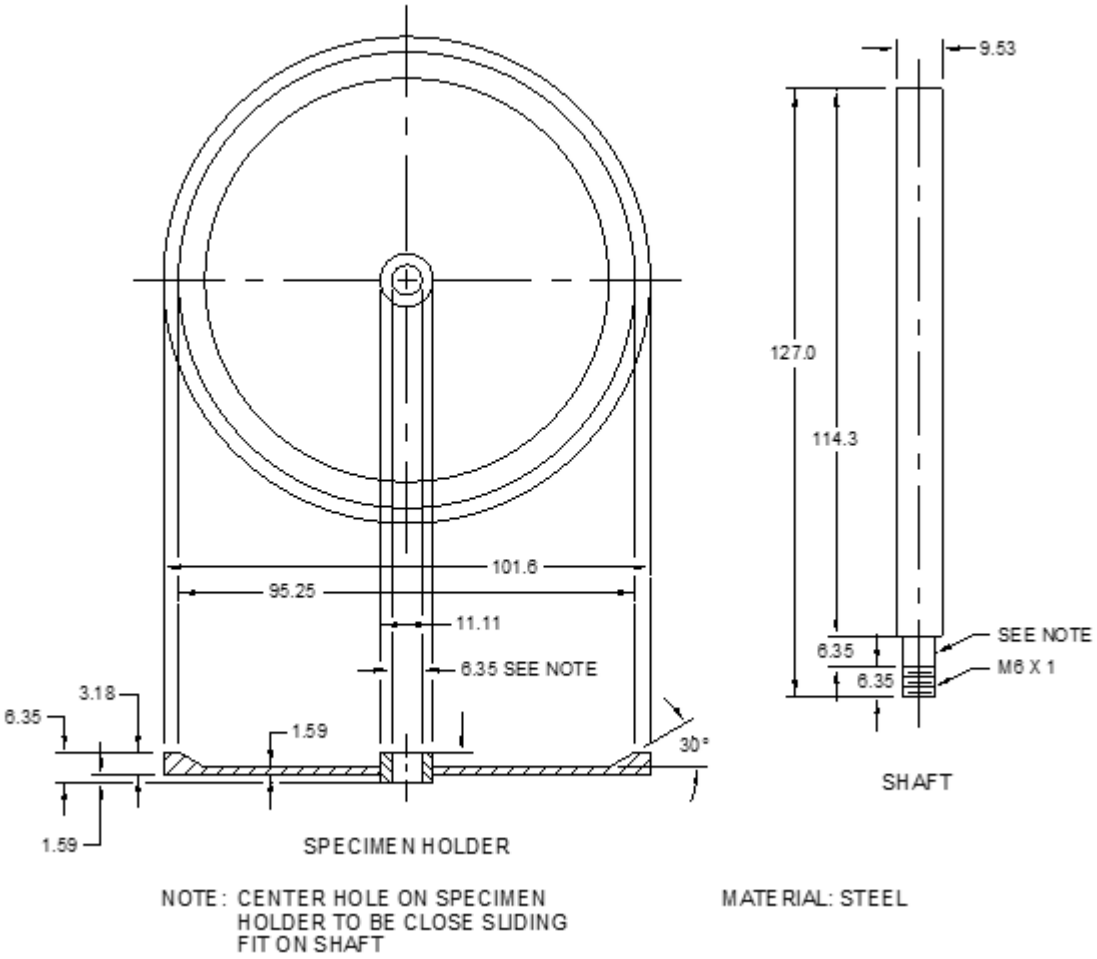


FIGURE 2. Apparatus for adhesiveness tests.

4.5.6.2 Procedure. The clean, dry disc shall be weighed and its concave side shall be filled with the grease. A straight edge shall be used to strike off the level. The disc and grease shall be weighed. The disc shall be placed with the grease side up on the end of the shaft in the oven at 66.1 °C (151 °F). The drip shield shall be placed around the disc. After 15 minutes soaking at the test temperature, the motor shall be started and ran for 15 minutes. The disc shall be removed, cooled, and weighed with the grease remaining thereon. The percent of grease remaining on the disc shall be calculated. This procedure shall be performed thrice and reported to the nearest 1 percent (the average percent remaining on the disc).

4.5.7 Aquatic biodegradability (for type II only). The aquatic biodegradability test shall be performed in accordance with OECD 301F. If one of the acceptable alternative test methods (listed in note 1 of table II) is used for biodegradability, report the standard method used.

4.5.8 Aquatic acute toxicity (for type II only). Aquatic acute toxicity testing shall be performed in accordance with OECD 202.

4.5.9 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3.1 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the grease using data provided by the manufacturer/distributor to the NMCPHC (see 3.3.1 and 6.5).

4.5.10 Prohibited materials. Prohibited materials shall be verified for conformance to 3.3.2 as required by the qualifying activity.

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4.5.11 Compatibility. Standard grease cups shall be filled with equal volumes of the grease being tested and currently qualified greases to ensure conformance to 3.6. The cup shall be filled by alternately adding 60 grams (2.11 ounces) of each of the two greases being tested, until the cup is full (approximately 454 grams [16 ounces]). The two greases shall be mixed by working the sample for 60 double strokes in accordance with the worked penetration procedure of ASTM D217. The mixture shall then stand undisturbed for 7 days at 18 to 29.5 °C (65 to 85 °F) and 50±10 percent relative humidity. The worked penetration shall be examined in accordance with table I, using the same sample of grease mixture for both tests. Small amounts of fresh grease may be added to the worker cup just prior to the worked stability test as needed to maintain a full cup for testing.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I or type II wire rope and exposed gear grease is intended to provide lubrication and corrosion protection for wire ropes and exposed gears. Type II provides an environmentally preferable formulation for applications where the potential for loss of lubricant to the environment is greater. This grease has unique physical and chemical requirements that are compatible for use across the Navy. Current stock of the grease may be used until depleted; future requisitions for the grease should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of grease (type I or II) (see 1.2).

c. Date of ordering and date of supply.

d. Quantity.

e. Sampling plan for the examination of filled containers and acceptable number of defective containers (see 4.3.3).

f. Packaging requirements (see 5.1).

g. When a Safety Data Sheet (SDS) is required (see 6.6).

h. Specific shelf-life requirements (see 6.7).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.18458 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements. Manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Definitions.

6.4.1 Lot. For purposes of sampling, a lot consists of all grease manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

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6.5 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

6.6 Safety data sheet. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract. The SDS should be included with each shipment of the material covered by this specification.

6.7 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, at a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-Life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https:/www.shelflife.dla.mil/.

6.8 Local environmental regulations. If local environmental regulations are more stringent than those listed in the specification, local regulations will supersede those listed in this specification.

6.9 Subject term (key word) listing.

Corrosion protection

Lubricant

Open gear

Preservation

Weather resistant

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – SH
Air Force – 03

Preparing activity:
Navy – SH
(Project 9150-2022-001)

Review activities:
Army – MR
Navy – AS, CG, SA
Air Force – 20, 68
DLA – GS, GS3, PS

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